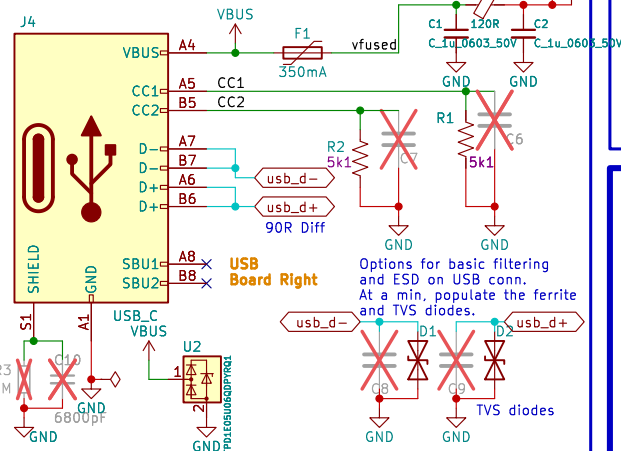
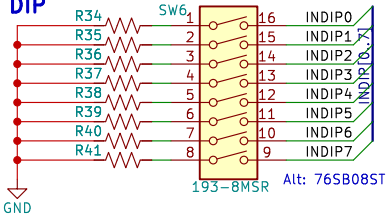


FPGA ASIC Teaching Board

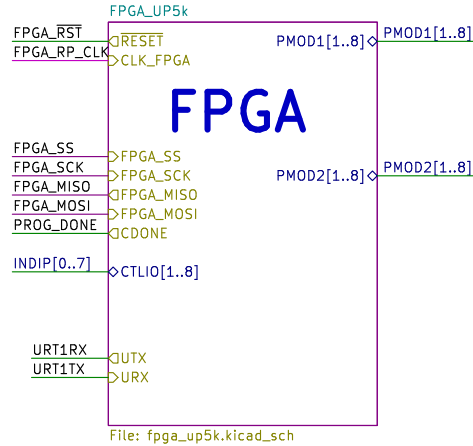
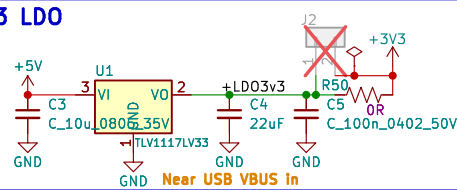
USB connector



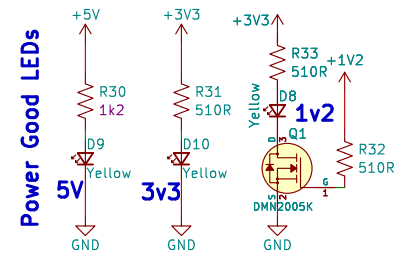
Input DIP



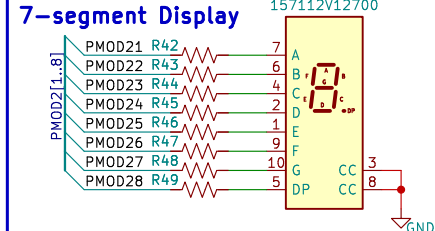
3V3 LDO



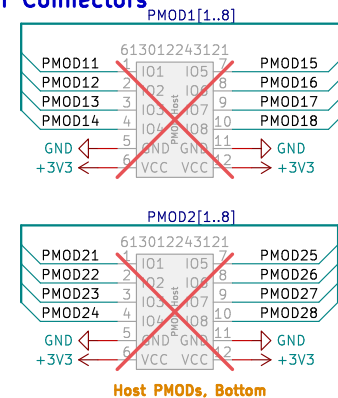
Indication



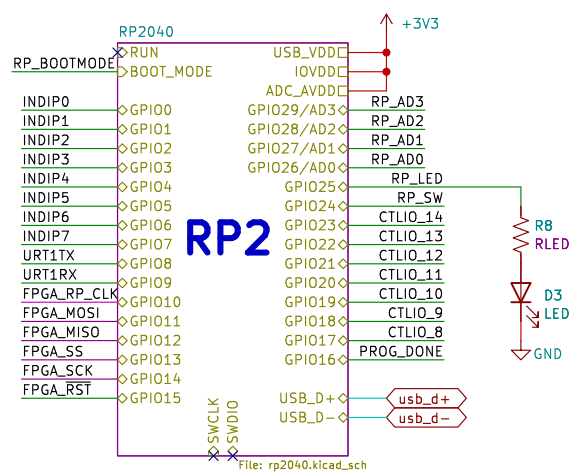
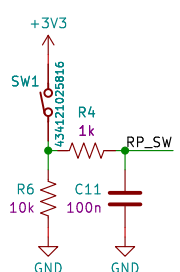
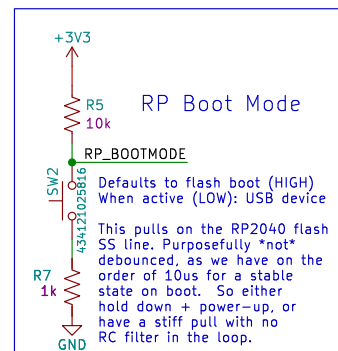
Output and External



User Connectors



RP2040 MCU



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Psychogenic Technologies for efabless
 Sheet: /
 File: ChipigniteExplain.kicad_sch
Title: Chipignite Explain
 Size: A4 Date: 2025-02-21
 KiCad E.D.A. 8.0.2 Rev: 1.1
 Id: 1/3

RP2040 and Support

IOVDD IOVDD

Logic supply, nominally 3v3.

BOOT_MODE QSPLSS

When held low on powerup, flash SS determines boot mode
(HIGH == flash boot, LOW == USB device)

USB_VDD supplies USB PHY, nominal 3v3. If IOVDD is 3v3, can share supply.

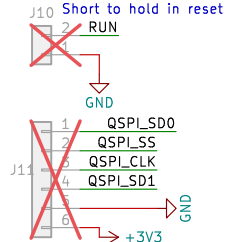
In fact, in this and many applications, IOVDD, USB_VDD and ADC_AVDD are all powered directly from a single 3v3 supply, with the 1v1 digital core being handle by on-board regulator.

VREG_VOUT: Int core regulator, 1.1V
Can supply DVDD
Place 1uF in/out bypass near pin.

RP2
Right of FPGA
Orient 1>
Top Right

CDONE up here

MOSI
MISO
CS
SCK
SPI CRAM



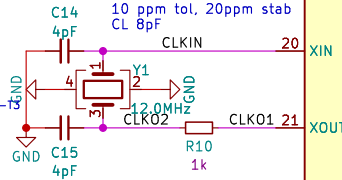
Flash program header
Note: should we replace 3v3 with RUN, to be able to reset/hold while updating flash?

RP2 xtal must be "exactly" 12MHz.
Recommended is ABM8-272-T3
Cload 10pF
Maybe go with ABM8W-12.0000MHZ-7-B1U-T3
Cload 7pF
for 10ppm rather than 30ppm stab.

Rule of thumb
C1, C2 = 2 * CL - 2 * Cstray
Using a stray cap of 5pF, gives
Cn = 4pF

Into:
 $CL = (C1 * C2) / (C1 + C2) + Cstray$
These Cn = 4pF give
CL = 7pF -- just what we need.

WE 830108206909:
CFPX-180 model
10 ppm tol, 20ppm stab
CL 8pF

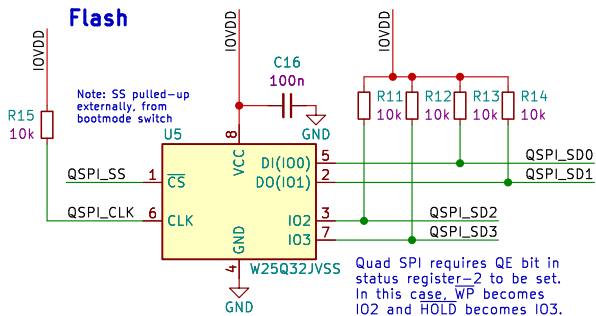


SWCLKD 24
SWCLK 25
SWDIO 25

TESTEN 19
Factory test mode: GND

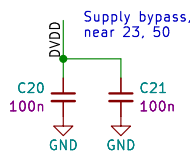
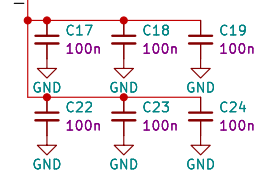
U4
RP2040

Flash



Quad SPI requires QE bit in status register-2 to be set. In this case, WP becomes IO2 and HOLD becomes IO3.

Supply bypass, place near 1, 10, 22, 33, 42, 49



GPIO	F1	F2	F3	F4	F5	F6	F7	F8	F9
0	SPI0 RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PI00	PI01		USB OVCUR DET
1	SPI0 CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PI00	PI01		USB VBUS DET
2	SPI0 SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PI00	PI01		USB VBUS EN
3	SPI0 TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PI00	PI01		USB OVCUR DET
4	SPI0 RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PI00	PI01		USB VBUS DET
5	SPI0 CSn	UART1 RX	I2C0 SCL	PWM2 B	SIO	PI00	PI01		USB VBUS EN
6	SPI0 SCK	UART1 CTS	I2C1 SDA	PWM3 A	SIO	PI00	PI01		USB OVCUR DET
7	SPI0 TX	UART1 RTS	I2C1 SCL	PWM3 B	SIO	PI00	PI01		USB VBUS DET
8	SPI1 RX	UART1 TX	I2C0 SDA	PWM4 A	SIO	PI00	PI01		USB VBUS EN
9	SPI1 CSn	UART1 RX	I2C0 SCL	PWM4 B	SIO	PI00	PI01		USB OVCUR DET
10	SPI1 SCK	UART1 CTS	I2C1 SDA	PWM5 A	SIO	PI00	PI01		USB VBUS DET
11	SPI1 TX	UART1 RTS	I2C1 SCL	PWM5 B	SIO	PI00	PI01		USB VBUS EN
12	SPI1 RX	UART0 TX	I2C0 SDA	PWM6 A	SIO	PI00	PI01		USB OVCUR DET
13	SPI1 CSn	UART0 RX	I2C0 SCL	PWM6 B	SIO	PI00	PI01		USB VBUS DET
14	SPI1 SCK	UART0 CTS	I2C1 SDA	PWM7 A	SIO	PI00	PI01		USB VBUS EN
15	SPI1 TX	UART0 RTS	I2C1 SCL	PWM7 B	SIO	PI00	PI01		USB OVCUR DET
16	SPI0 RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PI00	PI01		USB VBUS DET
17	SPI0 CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PI00	PI01		USB VBUS EN
18	SPI0 SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PI00	PI01		USB OVCUR DET
19	SPI0 TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PI00	PI01		USB VBUS DET
20	SPI0 RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PI00	PI01	CLOCK GPIN0	USB VBUS EN
21	SPI0 CSn	UART1 RX	I2C0 SCL	PWM2 B	SIO	PI00	PI01	CLOCK GPIN1	USB OVCUR DET
22	SPI0 SCK	UART1 CTS	I2C1 SDA	PWM3 A	SIO	PI00	PI01	CLOCK GPIN2	USB VBUS DET
23	SPI0 TX	UART1 RTS	I2C1 SCL	PWM3 B	SIO	PI00	PI01	CLOCK GPIN3	USB VBUS EN
24	SPI1 RX	UART1 TX	I2C0 SDA	PWM4 A	SIO	PI00	PI01	CLOCK GPIN4	USB OVCUR DET
25	SPI1 CSn	UART1 RX	I2C0 SCL	PWM4 B	SIO	PI00	PI01	CLOCK GPIN5	USB VBUS DET
26	SPI1 SCK	UART1 CTS	I2C1 SDA	PWM5 A	SIO	PI00	PI01		USB VBUS EN
27	SPI1 TX	UART1 RTS	I2C1 SCL	PWM5 B	SIO	PI00	PI01		USB OVCUR DET
28	SPI1 RX	UART0 TX	I2C0 SDA	PWM6 A	SIO	PI00	PI01		USB VBUS DET
29	SPI1 CSn	UART0 RX	I2C0 SCL	PWM6 B	SIO	PI00	PI01		USB VBUS EN

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Psychogenic Technologies

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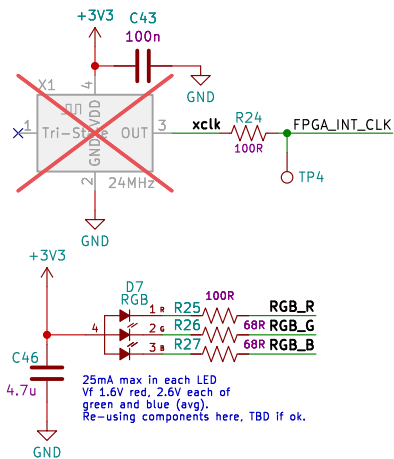
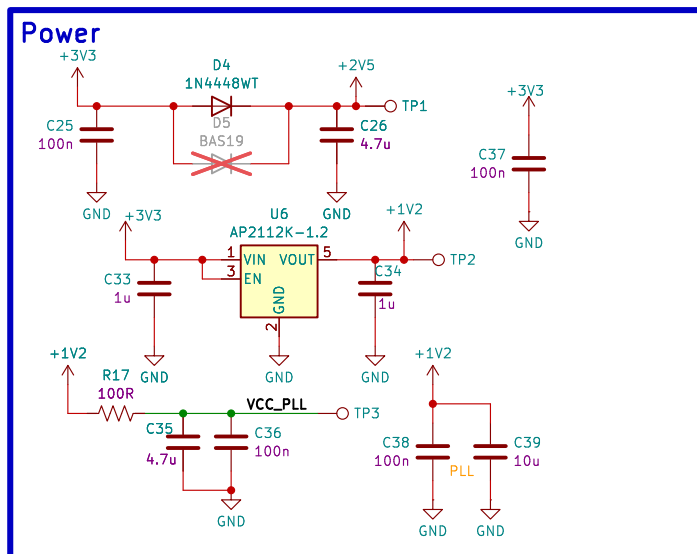
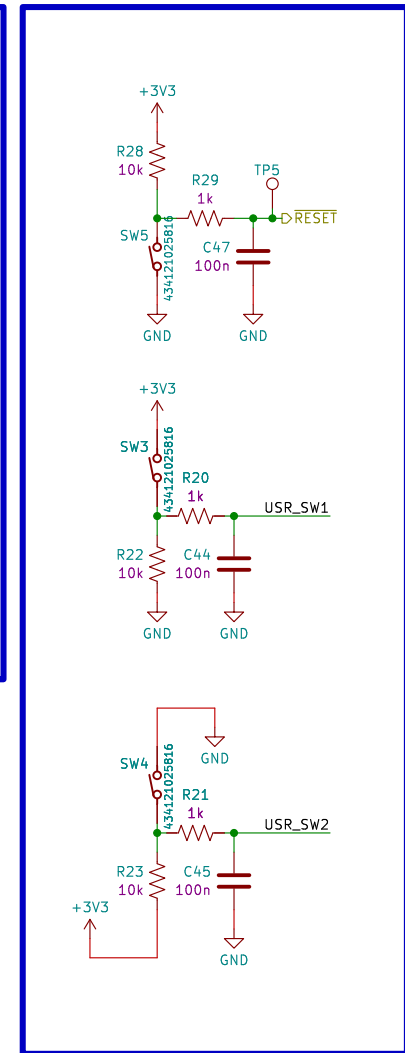
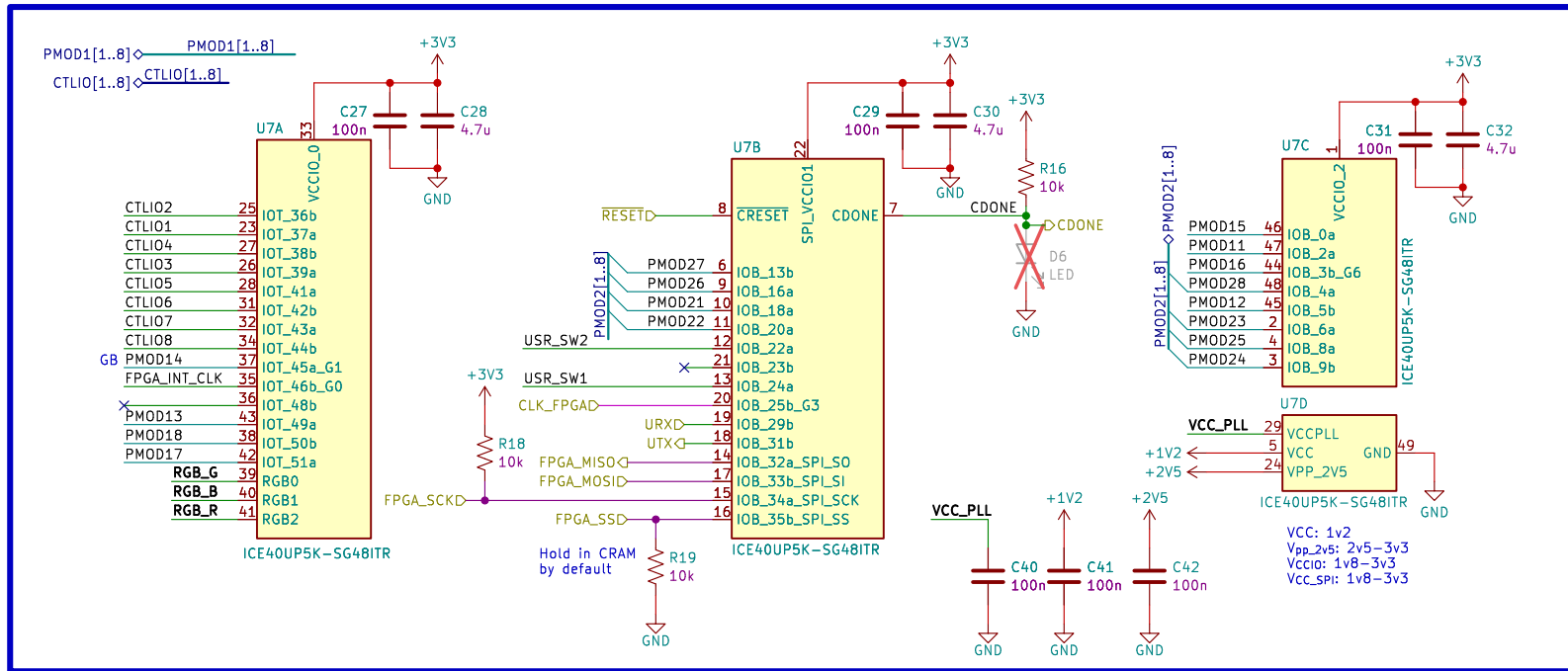
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Rev: 1.0

Id: 3/3

UP5K FPGA and Support



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Rev:
Id: 5/3